



Python Programming Fundamentals

CRUD Operations

Programming Fundamentals Team

SETU

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Outline

1. What is CRUD?
2. Create — Adding a Product
3. Read — Getting Data
4. Update — Modifying a Record
5. Delete — Removing a Record
6. CRUD in the Menu
7. Summary



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1.1 The four fundamental data operations

Letter	Meaning	HTTP verb
C	Create — add new data	POST
R	Read — retrieve data	GET
U	Update — modify existing data	PUT / PATCH
D	Delete — remove data	DELETE

CRUD is the backbone of databases, REST APIs, and any persistent system. Our Shop app implements CRUD on a list of Products.



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2. Create — Adding a Product

```
# In shop.py
def add_product(self, product: Product):
    # Check for duplicate ID
    if self.find_by_id(product.get_product_id()):
        raise ValueError(
            f"Product ID {product.get_product_id()} already exists"
        )
    self.__products.append(product)

# In driver.py
def create_product(shop: Shop):
    name = input("Name: ").strip()
    price = float(input("Price: €"))
    pid = int(input("ID: "))
    shop.add_product(Product(name, price, pid))
    print("Product created.")
```



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3.1 Retrieving one or all records

Get all

```
def get_all_products(self) -> list:  
    return list(self.__products) # return a copy
```

Get by name

```
def find_by_name(self, name: str):  
    for p in self.__products:  
        if p.get_name().lower() == name.strip().lower():  
            return p  
    return None
```

Get by ID

```
def find_by_id(self, pid: int):  
    for p in self.__products:  
        if p.get_product_id() == pid:
```



3.1 Retrieving one or all records

```
        return p  
    return None
```



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4. Update — Modifying a Record

```
# In shop.py
def update_product(self, pid: int,
                    new_name: str = None,
                    new_price: float = None) -> bool:
    p = self.find_by_id(pid)
    if p is None:
        return False
    if new_name is not None:
        p.set_name(new_name)
    if new_price is not None:
        p.set_price(new_price)
    return True
```

```
# In driver.py
def update_product_menu(shop: Shop):
    pid = int(input("Product ID to update: "))
```



4. Update — Modifying a Record

```
new_price = float(input("New price (or 0 to skip): "))
updated = shop.update_product(
    pid,
    new_price=new_price if new_price > 0 else None
)
print("Updated." if updated else "Not found.")
```



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5. Delete — Removing a Record

In shop.py

```
def delete_product(self, pid: int) -> bool:
    p = self.find_by_id(pid)
    if p:
        self.__products.remove(p)
        return True
    return False
```

In driver.py

```
def delete_product_menu(shop: Shop):
    pid = int(input("Product ID to delete: "))
    p = shop.find_by_id(pid)
    if p is None:
        print("Not found.")
        return
    confirm = input(f"Delete '{p.get_name()}'? (y/n): ")
```



5. Delete — Removing a Record

```
if confirm.lower() == "y":
    shop.delete_product(pid)
    print("Deleted.")
else:
    print("Cancelled.")
```



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6. CRUD in the Menu

```
def show_menu():
    print("\n=== Shop CRUD ===")
    print("1. Create (add product)")
    print("2. Read all (list products)")
    print("3. Read one (find by ID)")
    print("4. Update (change price)")
    print("5. Delete (remove product)")
    print("0. Exit")

def main():
    shop = Shop("My Shop")
    while True:
        show_menu()
        match input("Choice: ").strip():
            case "1": create_product(shop)
            case "2": shop.display_all()
```



6. CRUD in the Menu

```
case "3":
    pid = int(input("ID: "))
    p = shop.find_by_id(pid)
    print(p if p else "Not found.")
case "4": update_product_menu(shop)
case "5": delete_product_menu(shop)
case "0": break
```



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7. Summary

CRUD	Python list op	Our method
Create	<code>append()</code>	<code>add_product()</code>
Read all	for loop	<code>get_all_products()</code>
Read one	linear search	<code>find_by_id()</code>
Update	find + set attr	<code>update_product()</code>
Delete	<code>remove()</code>	<code>delete_product()</code>



Thanks for Watching - Any questions?

